

Primal Dual — Shortest Path

Shortest path and its Dual

$$\min: c^T f$$

$$\text{st: } Af = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}^T$$

$$f \geq 0$$

$$\max: p_s$$

$$\text{st: } p_i - p_j \leq c_{ij}, \quad \forall (i,j) \in E$$

$$p_t = 0$$

$$p \text{ free.}$$

Admissible columns (arcs) for a dual feasible p

$$= J = \left\{ (i,j) \in E : p_i - p_j = c_{ij} \right\}$$

Restricted Primal (RP) and its Dual (DRP)

$$\text{RP: } \min: \sum x_i^a$$

$$\text{st: } Af + x^a = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

$$f_{ij} \geq 0 \quad \forall (i,j)$$

$$f_{ij} = 0 \quad \forall (i,j) \in J^c$$

$$x^a \geq 0$$

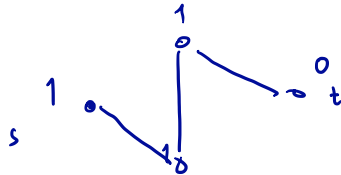
$$\rightarrow \text{DRP: } \begin{aligned} \max: & p_s \\ \text{st: } & p_i - p_j \leq 0 \quad \forall (i,j) \in J \\ & p_i \leq 1 \\ & p_t = 0 \\ & p \text{ free.} \end{aligned}$$

max: p_5

$$\text{st: } \underline{p_i - p_j} \leq 0 \quad \forall (i, j) \in \underline{J}$$

$$p_i \leq 1$$

$$\bar{P} = \begin{cases} 1 & \forall i \text{ reachable from } s \text{ using arcs in } I \\ 0 & \forall i \text{ from which } t \text{ is reachable " " } \\ 1 & \text{all other nodes} \end{cases}$$



$$\theta = \min_{\text{over } (c_{ij}) \in J^c} \left\{ c_{ij} - (p_i - p_j) \right\}$$

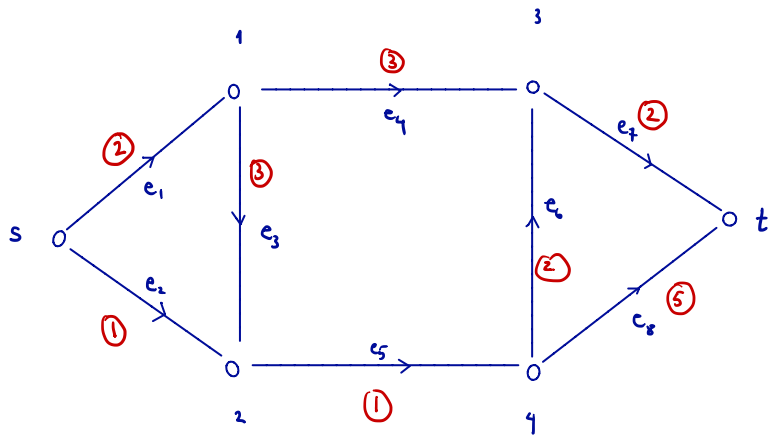
st: $\bar{p}_i - \bar{p}_j > 0$

The argmin $(\hat{c}_i, \hat{f}_j)$ becomes tight
for $p_{\text{new}} = p + \theta \bar{p}$

⇒ New edge in reachability problem
which is DRP.

As soon as there is a path in J from s to t

Shortest Path via P-D Example - 1/6



max: P_s

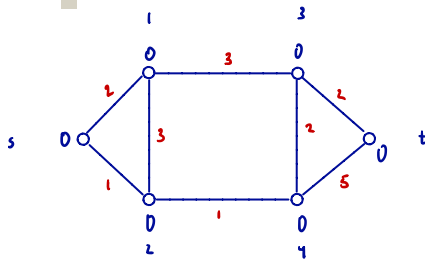
st: $P_i - P_j \leq c_{ij} \quad \forall (i,j) \in E$

$P_t = 0$

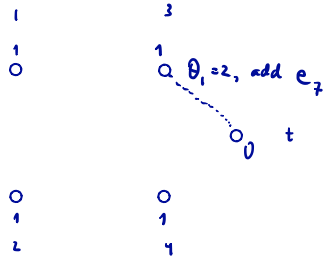
P free

$p=0$ is a
dual feasible point,
hence initialize P-D
here

Shortest Path via P-D Example - 2/6



$\Rightarrow$



$$D: p = (0, 0, 0, 0, 0, 0)$$

$$J = \{(i, j) : p_i - p_j = c_{ij}\} = \emptyset$$

DRP

$$\text{DRP: max: } p_s$$

$$\text{st: } p_i \leq 1 \quad \forall i$$

$$p_t = 0$$

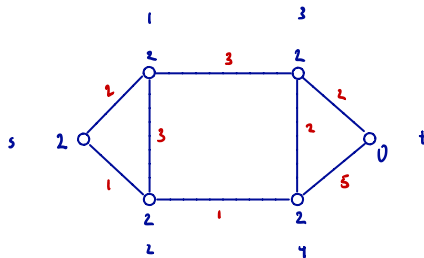
$$\bar{p} = (1, 1, 1, 1, 1, 0)$$

$$\Rightarrow \theta = \min_{(i, j) \in J} \{c_{ij} - (p_i - p_j)\}$$

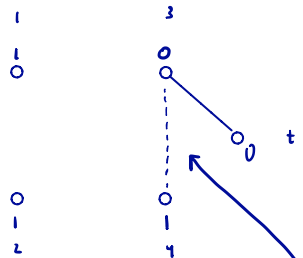
$$\text{st: } \bar{p}_i - \bar{p}_j > 0$$

$$= 2 - 0 = 2 \quad \text{for edge } (3, t)$$

Shortest Path via P-D Example - 3/6



$\Rightarrow$



$$D: p = (2, 2, 2, 2, 0) = p_{old} + \theta \bar{p}$$

$$J = \{(i, j) : p_i - p_j = c_{ij}\} = \{7\}$$

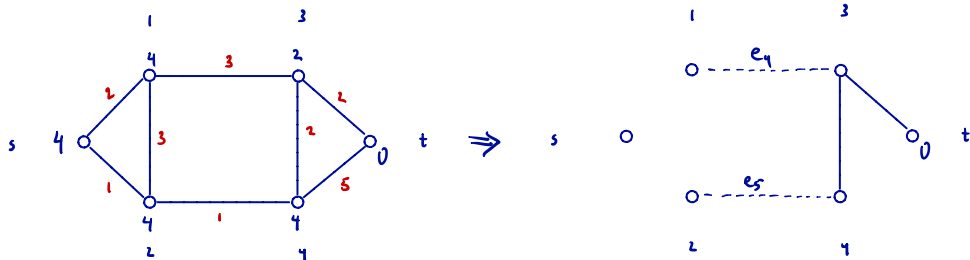
$$DRP: \bar{p} = \begin{cases} 1 & \text{nodes reachable from } s \\ 0 & \text{nodes from which } t \text{ reachable} \\ 1 & \text{other} \end{cases}$$

$$\bar{p} = (1, 1, 1, 0, 1, 0)$$

$$\theta = \min_{(i, j) \in J} \{c_{ij} - (p_i - p_j)\} \\ \text{s.t. } \bar{p}_i - \bar{p}_j > 0$$

$$\Rightarrow \theta = 2, \text{ adds edge } e_6$$

Shortest Path via P-D Example - 4/6



$$D: p = (4, 4, 4, 2, 4, 0) = p_{old} + \theta \bar{p}$$

$$J = \{7, 6\}$$

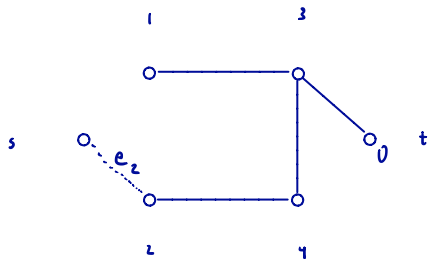
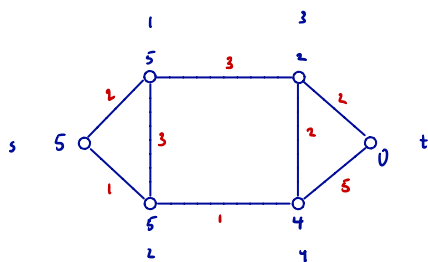
$$\bar{p} = (1, 1, 1, 0, 0, 0)$$

$$\theta = \min_{(i,j) \in J} \{c_{ij} - (p_i - p_j)\}$$

s.t. $\bar{p}_i - \bar{p}_j > 0$

$$\Rightarrow \theta = 1, \text{ adds edge } e_4, e_5$$

Shortest Path via P-D Example - 5/6



$$D: p = (5, 5, 5, 2, 4) = p_{old} + \theta \bar{p}$$

$$J = \{7, 6, 5, 4\}$$

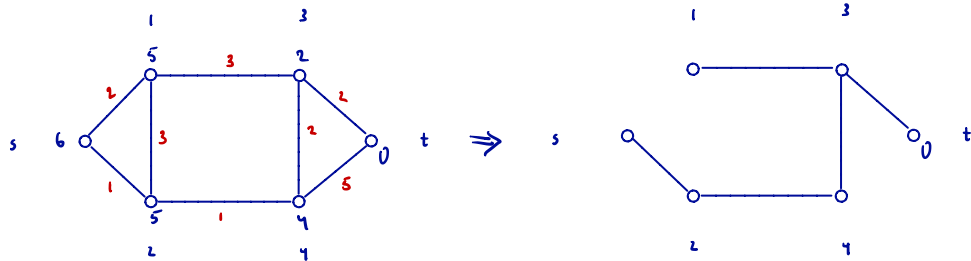
$$\bar{p} = (1, 0, 0, 0, 0, 0)$$

$$\theta = \min_{(i,j) \in J} \{c_{ij} - (p_i - p_j)\}$$

s.t. $\bar{p}_i - \bar{p}_j > 0$

$$\Rightarrow \theta = 1, \text{ adds edge } e_2$$

Shortest Path via P-D Example - 6/6



$$D: p = (6, 5, 2, 4, 0) = p_{all} + \theta \bar{p}$$

$$J = \{7, 6, 5, 4, 2\}$$

$$\bar{p} = (0, 0, 0, 0, 0, 0)$$

$\Rightarrow$ optimal solution

Discussion

$$W = \{i : t \text{ reachable from } i \text{ using edges in } J\}$$
$$= \{i : \underline{\bar{p}_i} = 0\}$$

p_i remains fixed once $i \in W$.

Can show: $\forall i \in W$, $p_i = \text{value of shortest path from } i \text{ to } t$.

↳ Dijkstra's Alg.